You are listening to the Common Descent Podcast. This is an interesting topic because there's a bunch of stuff to discuss with the deep sea. We have the things that live down there, what conditions they have to live under all that water, and the history, fossil history, and how things got to be the way they are down there, how the organisms got to be the way they are down there. This is also just the deep sea, and that's sort of the broad term, is one of the most expansive habitats on the planet.

Exactly. Part of the reason there's so much to cover in this topic is just because it is the largest expanse of our planet of varying habitats. Not as densely habitated as other places necessarily, but wide and diverse. There's a lot of different, interesting, and wholly unique habitats in the deep sea. Can't wait to hear about it. Now as usual, we are discussing this topic because it was requested, this episode was requested by Ciaran, Hobart's Obsession, Alejo, and Joshua, a couple of which who suggested some of the habitats we'll be discussing.

Now before we get into the episode, we have some announcements. Our first announcement is that we have a Patreon, and if you sign up on our Patreon to support us and take advantage of all the extra goodies we put on Patreon, at a certain level and up, we will shout your name out here on the podcast. What's that sound like? It sounds somewhat like this. Welcome our new patrons, Tane, Amrita, Jackson, Michael, Claire, and Tim. Welcome everybody. Hey, those patrons don't just get to enjoy their names shouted out, they also get stuff like bonus audios and a recent thing that we're doing more commonly, live streams with our patrons where they can interact with us directly. So hey, if you want to support the podcast, support the science education-y stuff that we do, consider subscribing to Patreon.

Yeah, Patreon supports us by funding everything we do. And speaking of supporting us, we recently got some Christmas packages in the mail. Oh yeah, we did. We have a physical mailing address now, which you can find in the episode description. We're not saying you have to send us holiday presents. No! Of course not. But we got some. We did get some and we are available. We have lots of space. Yes, absolutely. And these were really cool presents, so.

Also speaking of interacting with us, the Q&A forum, the end of the year Q&A is still up for a little bit. Yeah, the Q&A forum ends on December 15th. It will close. So this episode comes out three days before that. Yep, so if you want to submit questions that we will answer on the end of the year Q&A and will go up right before New Year's, go to that forum, link in the description and on the blog and on our social media, and submit your questions. You have just a few days left.

Yep, and if you're not listening to this episode until after December 15th, listen to the Q&A when it comes out at the end of the year, and then submit a question next time. Yes, exactly. And in addition to that extra stuff, we've recently done a kind of impromptu silver screen science. Yeah, we talked about a movie kind of. Yeah, we talked about a movie promo that came out and just begged to be discussed. So if you haven't already, check out our short-ish episode of Silver Screen Science about the Jurassic World Dominion prologue/trailer that came out recently.

It was a lot of fun, actually. We had many things to say about it. And with that, we can wrap up our announcements to move on to the first official section of the episode, the news! Every episode we like to gather up recent paleo-evolutionary earth science news, things that keep us all up to date in the goings-on of research. David, what's the news for today? I'm going to start with a news of a dinosaur that has enamored the internet.

A dinosaur people have been talking about? People have been talking about a dina, and not even in a movie. This is a new ankylosaur with a really weird tail, and it's a weird tail even for an ankylosaur. Even compared to the group known for weird tails. This is research by Sergio Soto-Acuna et al. in the journal Nature, and we will link on the blog post, as always there is a blog post after the episode, to an article in National Geographic by Michael Greshko.

Armored dinosaurs, sort of the colloquial name that encompasses two groups, ankylosaurs, the tank dinosaurs, episode 69, and stegosaurs, like stegosaurus, episode pending, both groups known for armor, right, osteoderms, plating and stuff on the body, and oftentimes tail weapons. Look at this. Stegosaurs often have spiky tails, and ankylosaurs, at least some of them, have clubs on their tail. Now, armored dinosaurs are rare in the southern continents, in Gondwana. They're only a handful definitively known. This new dinosaur, in addition to being shaped weird, is also from the southern continents.

It's from Gondwana. This one comes from southern Chile, from the late Cretaceous, around 75 to 72 million years ago, which means it's a rare find for this part of the world. It has been given a new genus and species name, Stegoros ellengasen. Let me tell you about all the things that's cool about it. I'm ready. For one, it's known from a mostly complete skeleton. The article said around 80% complete, semi-articulated, which means it was at least somewhat still in the shape of an animal skeleton.

That's cool. A lot of dinosaurs go not huge, about two meters long, so six, seven feet, would only have been about two feet tall. This was a little dinosaur. Totally fit in your backyard. Absolutely. As it was an ankylosaur, covered in armor, so just bumpy platyosteoderms around its body, but the thing everyone's talking about is its tail. Some ankylosaurs just have tails, normal-looking tails. Then they're fused and stiff towards the end, and in the later ankylosaurs, certain ones had clubs, just big, knobby clubs at the end of the tail.

Like a mace. Like a mace. This one has a series of flat structures, I saw them described as like fronds, or you might think of like, I don't know, shingles in a row, seven pairs of flat osteoderms that stick out to either side, creating what looks like an axe blade, or even a sword blade. Yeah, it makes me think of those South American wooden swords with the obsidian blades embedded in the side. In fact, the researchers named it after that.

Yes! They named the researchers called the structure a makwa-huatl, after the obsidian bladed wooden clubs used by Aztec people. Absolutely, you are 100% correct. It's in much the same way that ankylosaur tail clubs are thought to have been used as bludgeoning weapons, either in, you know, fighting other members of your species or warding off predators. This may have functioned kind of like a slashing weapon, that it was a flattened version of the tail weapon, which we have not seen in any other dinosaur or any other animal.

Good point. It's kind of like if a swordfish nose was on the back end of an animal. Yeah, like a sawfish tail. Yeah, like a sawfish, or something like that. The species epithet, so elangassan, is named for an armored beast in the mythology of certain Patagonian peoples, so just a very cool structure on a dinosaur. Yeah, it's fascinating because, like you said, it's already a group with weird tails, and this just stands out for not only being weird among them, but just weird among tail structures as far as we know in the animal kingdom.

But also, it makes complete sense to me, like, even if it's not edged, even if the edge was not razor sharp in life. Not bladed. But, if I'm hitting you with something that weighs a lot, the smaller the surface area I'm hitting you with, the more force I'm imparting upon you when I hit you with it. So like, it's still going to do a lot of damage even if it's not sharp sharp. Yeah, and if you're little, you're right at like ankle and shin slicing level.

Just take out that Achilles tendon. I think. But, what's really cool is that the interesting things about this dinosaur don't stop at the tail. There's a bunch of scientifically, like somewhat more technical, fascinating things. For one, again, found in the southern continents in what used to be Gondwana, which is unusual for these dinosaurs, and the skull, teeth, and tail are very much like other ankylosaurs, but the leg bones and the pelvis, the hip bones, look a lot like stegosaurs.

So it has a few features that seem to be early features of the group that ankylosaurs eventually lost and stegosaurs retained, since those two groups are closely related. This is an ankylosaur that still has some stegosaur-like features. The researchers did a phylogenetic comparison that found that Stegoros is an ankylosaur, but part of an early branch, the earliest branch of ankylosaur. So a very early basal, as we would call it, split from ankylosaur, which explains why it still has some of these ancestral features.

And it's not alone. It is most closely related to an Australian ankylosaur named Cumbarasaurus, and an Antarctica ankylosaur named Antarcticopelta. Altogether, this study names that group Paranchylosauria, an early branching group of ankylosaur. - I like that this is kind of like, it makes me think of lemurs, to where they have features of more, things that we don't think of as common primate features because they were more common in ancient groups, and they've retained them, but they are not ancient themselves.

They just retain some of these features because they're not part of the main group that evolved away from it. - Yeah, this group has some things in common with stegosaurs, despite living many tens of millions of years after stegosaurs dwindled. - That's, that's so interesting. - Now there are a couple of things that this says, not about Stegoros, but implications for other dinosaurs. For one thing, as a researcher has quoted in the National Geographic article pointing out that the pelvis of stegosaurs is one of the things that defines stegosaurs.

- Oops. - And so this might force us to adjust what features we use for identifying stegosaurs versus ankylosaurs, because we didn't know ankylosaurs also could have these, that structure. - Yeah, we didn't know they could have the same hips. - And the authors point out that Antarctica pelta, the ankylosaur from Antarctica, not known from very many bones, just bits and pieces, so we don't have a really good sense of what it looks like, has some features in its tail bones that are similar to this new one.

- Oh. - So it might be that stegoros isn't the only ankylosaur we know of with this obsidian blade tail. And Antarctica pelta, which is twice the size of stegoros, possibly also had this structure. And the authors point out that there are, in the National Geographic article they say, they know where more stegoros fossils are. It's not just this skeleton, they know there's more out there. So we might actually get the chance to study a bit more in depth this particular species of dinosaur.

- Oh, that's so exciting. - My favorite thing about this is that I saw it in the news and I went, oh, cool, a blade tail, definitely I'll talk about that. And then I read and it turns out that there's so much interesting stuff about this dinosaur besides its obsidian blade tail.

- Yeah, the blade tail is just the tip of a weird ankylosaur iceberg. - Yes, it is, in so many ways. - That's so interesting.

I love just it's a bundle of weirdness and the fact that there's more to come, very likely more to come in future expeditions is so exciting. And wait, oh, tell me more, a well, a well, a well, a huh. - Well, I can't tell you more about that, but I can tell you about a Triassic pseudosuchian with feeding habits different from what it looked like it would have had. - Okay, this is a weird croc thing.

- Yeah. This is research by Jordan Bestwick et al in the anatomical record and the article is a press release by the University of Birmingham. So pseudosuchians are a group among archosaurs. They are the group that includes crocodilians, but also all the not croc, but croc related. - All the croc cousins, the extended croc family tree. - So this is the group that's related more closely to crocodilians than they are to the other archosaurs. So everything on that side of the wedge.

In the Triassic, they were the big diverse group, like they were really pretty dominant on land before dinosaurs took on all those dominant roles. But there are lots of instances of very dinosaur "shaped" pseudosuchians during the Triassic, like filling very similar ecological niches and in very similar body shapes that we would later see in dominant dinosaurs. - Yes, small two-legged running ones, big predatory ones, big armored ones, all sorts of dino shapes. - Absolutely. This is one in particular named Ephigea o'Keefe, which was a medium to smaller sized, they said like gazelle sized.

That was lightweight, long thin-limbed, longish neck, large eyes, beak-like mouth, seemed to be standing up on its hind limbs, very ostrich-shaped. Very similar to your Struthiomimoths and Ornithomimoths, your ostrich-like dinosaurs, your ostrich mimic dinosaurs. - The ones the ostriches are mimicking. - Yes, exactly. This was kind of interpreted as another one of those "ostrich mimics" and slotted into there. Assume that it was probably living a similar way, herbivory, eating plants, pecking at the ground like ostriches and it's often thought that the ostrich mimics were probably doing.

It was found in North America, dates back to, the fossil dates to about 205 million years old. But the initial specimen that this description came from was not really well preserved. It wasn't complete. It had a skull and the skull in particular was deformed. - Gotcha. So it changed my time. - Exactly. Just mushed a little bit. So that was kind of the picture for a while. This research is taking a closer look at re-examining the specimen for its eating behavior, particularly.

And to basically see, is it really an ostrich mimic? Like does it actually, is it actually convergent with ostriches and the ostrich-like dinosaurs? - Right. It's shaped like an ostrich, but did it live like an ostrich? - Precisely. They did this using CT scans to get a much more accurate reconstruction of the skull, get a detailed look at its overall structure. They found a couple of things already that stood out from the initial reconstruction. The brain case was more rounded and more bulbous than it initially thought.

So a slightly different shape and sounded larger. The upper and lower jaws were also curved in a way that was not initially noticed. - Interesting. They said it was unlike an ostrich bill in that way. These kind of fit together more like shears than an ostrich bill. They then analyzed the skull from a couple of different perspectives. One was to see how much it actually seemed to be convergent by comparing it with the cranium of modern ostrich, ornithomimus and alligator skull.

- All right. So ostrich, ostrich-like dinosaur, and then just alligator. - Just living relatives. So that's why that one's there. So that they could actually see what degree does the convergence actually seem to hold up when the skull is looked at in detail. Upon analysis, they found limited functional convergence between effigia and the ornithomimus and ostrich skulls. - So not all that similar. - That don't seem to be functionally similar, just maybe superficially to us similar.

They then also looked at modeling the forces that the skull would need to withstand, need to undergo for that pecking ostrich-like feeding behavior to actually see, could they do that? And according to the analysis, when they calculated it, no, they said it probably would shatter. - So effigia would break itself. - If it were pecking at the ground like an ostrich, indicating that it was probably actually a more nimble feeder, nipping at things here and there, nipping at softer tips of plants with the tip of their jaw, that's where the force seemed to be most effective.

So with this study there, reinterpreting effigia as a specialist herbivore, taking in softer plant material, being selective, being kind of a picky eater potentially, maybe young shoots or new ferns kind of stuff, softer things. Which stands out because that is not a feeding style really known from the Triassic herbivores. So it could make effigia, probably not the only one doing this, but currently one of the only ones known with this sort of specialized feeding habit during the Triassic.

- That's pretty cool. I really enjoy studies like this because when I was a kid, you learn about how do we tell what fossil animals were eating, and it's very easy, especially when you're young, for that to be communicated as they ate meat or they ate plants. And then you might even get into more detail and learn the difference between browsers and grazers. But I really like that especially with modern techniques, with CT scanning and with biomechanical analyses of bones, we can get it even more detailed.

Not just what you were eating, but how you were eating it and what specific kinds of foods were you targeting. That's a level of detail that is really exciting to be able to examine in the fossil record. - It's allowing us to get into very specific niche partitioning among an ancient group of herbivores or predators. - And that tells us a bunch about the ancient ecosystem. You mentioned that this is just over 200 million years old, which puts it right at the end of the Triassic, which makes me wonder, was this something that was new?

Should it just showing up right towards the end of the period? Very cool. Well, for my next news, I am going to move us more in the direction of this episode's main topic, and we're gonna move into the ocean. But this is also a news about one organism being inside another organism, but not about feeding. - Oh, like a chest burster. - Exactly. - Okay. - This is about ancient xenomorphs. It's not. And this is research about shrimps hiding inside a clam.

- Oh, like a chest burster. - This article is by, the paper is by Russell Bicknell et al. in the journal, "Paleogeography, paleoclimatology, paleoecology, P3," for the lazy scientist. - For me. - And we will link to an article, another one in National Geographic, this one by Rebecca Zombach. Here's a vocabulary term that I learned today, "Inquilinism describes the phenomenon of an organism living inside another organism." - Inquilinism? - Inquilinism. - Cool. - This happens a ton.

I mean, we have organisms living inside our personal organisms. - So we did a whole episode about these things that like to live inside of other things. - Yeah, we sure, we

did a whole 102 about parasites, and indeed, the article points out that most of the records of organism interactions in the fossil record are either predation, right, bite marks and gut contents, or parasitism, parasites living inside the bodies of other things. But there's not much evidence of inquilinism of the various forms that we see today of organisms finding various reasons to live inside something else.

- Makes me think of the pearl fish, which goes inside the anus of the sea cucumber. - Yeah, just a place to live. - Yeah, it just hides in there. - This research describes a fossil clam from the Cretaceous around 100 million years ago in Queensland, Australia. It was discovered in 2016. The fossil clam itself is an eno-ceramid clam, common type of clam during that time if I remember correctly, that is about 25 centimetres wide, or 10 inches, which is a good-sized clam.

- That's a hefty clam. - The interior is filled with sediment, so it's all infilled with mud and silt. And within the silt, there are three shrimp. - Ooh, I didn't know there was more than one. - There are three swimming shrimps, so this is a swimming type of shrimp. Each of those is about three centimetres long, so a little over an inch long, tiny little shrimps. As you can see, hiding inside clams is a thing that shrimps sometimes do.

They go in there to seek shelter, but we don't have a lot of evidence of this kind of behavior in the fossil record, which makes this exciting. The authors point out that the shrimps were probably alive when they were buried, when the clam became infilled, because shrimps are relatively delicate-bodied animals, so they tend to break apart when they're moved. So if they had gotten washed in there or something, you might expect them to have broken down along the way.

These were in there, and then they got buried alive. - Nice. - Sad little shrimps. It's possible that they ran in there for nesting or for molting, you know, parts where you might be vulnerable. But there's no evidence inside the clam of any of those things, the author said, which I assume means that they didn't find evidence of a nest or eggs or shed skin or anything like that. So it's possible, and the hypothesis they're leaning towards is that the shrimps ran in there just to hide for shelter.

They found a nice big spacious clam and ran inside. In support of this, there is another clam found nearby that they didn't describe in detail in this study, a larger clam, and inside that clam, they found around 30 little fish fossilized inside that clam, which may have been doing a very similar thing. They may have inquilizized that clam. I can make up words too. And indeed the fact that they're found so close to each other not only supports that these clams were potentially used as shelter, but that they may have been hiding from the same thing.

- Oh, good point. - That it may have, there may have been a storm or some sort of environmental disturbance that sent animals scurrying for shelter or a predator swam by that sent a bunch of things scurrying for shelter. The authors point out that there's no evidence of a coral reef in the area. So it may be that there weren't convenient coral structures to hide under. So clams were the best option for these animals. - That's really neat.

Like the concept of them hiding clams is cool in and of itself. - The fact that we have two different clams with two different groups of different animals. - Yeah. - That's very, it makes me wonder if this site was just particularly good at preserving the qualities of the site and that's why we have two here, but we don't have like tons of other examples. - Yeah, yeah. Well, and you gotta imagine if they're just hiding and for a short period of

time, then there's not a lot of opportunity for that to be fossilized.

'Cause they're just gonna leave. So it might even be, and I didn't read through the whole paper. So I don't know if they discussed this, but they do mention in the article that it could be they were hiding from a storm. It could very well be that the event that sent them hiding also buried and fossilized the clams. - That it was a perfect scenario to get them in there and then keep them in there. - Yeah, forever.

- I have to say, I was picturing with the shrimps locked inside the clam and buried alive, just Kill Bill, what was it, the four-inch punch? - Yeah, yeah, yeah. Well, now I'm thinking of like mantis shrimps. - Yeah, yeah, yeah. - Just trying to, yeah, they're not as good at it as Uma Thurman. - No, no. - Apparently. - They needed an Uma Thurman shrimp. - The authors point out that this is the oldest known record, the oldest fossil of enquilinism between decapods and bivalves.

So decapods being things that include shrimp and bivalves, clams, oysters, et cetera. There are older records of animals inside other animals. All the way back to the Ordovician, we have records of animals hiding inside nautiloid and ammonite shells. And we recently discussed hermit worms in a recent news. And the authors point out that this is exciting because as we get more information about this behavior, we can understand how that behavior evolved and shifted over time. So like this is one of those kinds of cool topics that I would love, and this might not be something we can do yet, maybe in another hundred years or so, to see just a list of all the examples through time to see if we can get like, what organisms were the best hiding spots?

- Oh yeah. Like what was the Paleozoic's best hiding spot? And in the Cretaceous, was it clams? Was that the best place to hide? - Yeah, whose bodies were the go-to spots for hide and seek? - Yes, exactly. And who were the seekers? - Yes. - At a hundred million years ago, you know, plesiosaurs and sharks and ichthyosaurs, monsters. - That's fantastic. Well, speaking of sea monsters like those, my last bit of news is about an ichthyosaur with a mean set of chompers.

- Okay. - This is research by Durli Cortes et al. in the Journal of Systematic Paleontology, and the article is a press release in SciNews. This is another re-description of a previously known specimen. This is a holotype specimen from the ichthyosaur group Ophthalmosauridae. This is Platypterygius sachycarum. It's from the early Cretaceous, 130 million years ago, found in today's Columbia. It is known from a well-preserved, meter-long skull. - You said meter-long, so ichthyosaurs episode 116, the dolphin-shaped reptiles, shark-shaped, you know, fish-shaped, fish lizards.

You said a meter-long, and I was like, okay, that's not too big, skull. - Yeah. - Oh, okay. - This is one of the hefty ones. - That's a big ichthyosaur. - This research is taking a closer look at it, its features, and putting it into a new phylogenetic context. They're looking at where it fits within its group and related groups. First and foremost, this analysis agreed it is a unique species. That's the first thing it said, which I appreciated.

- Okay. - It's remaining its own species, but there are some new characteristics noted in this analysis in the nariel region, the mandible, and the dentition is what's been making a lot of the news. - Oh, nose, jaws, and teeth. - The teeth of this ichthyosaur are notable for having separate morphologies, different shapes. - All right. - In the jaw. - Within the same mouth? - Yes, within the same mouth. - Hey, we talked about that all the way back in episode 88 about teeth.

- Yes. - Heterodont dentition. - So this is a heterodont ichthyosaur. Most ichthyosaurs

are known from uniform, typically conical, sharp, often small teeth. - Right. Kind of like certain crocodilians. - Yeah, like a gharial. - Or even like dolphins. - Yes. Very dolphin-esque, good for catching small to maybe medium-ish prey. That's what most of them are known for. This has both cutting... This has piercing, cutting, and crushing teeth. - Ooh. - And they are large enough to indicate that this was eating large, very likely vertebrate prey.

- Ooh. - So this was a macro carnivore. It was taking down big, hefty prey. - An all-purpose carnivore. - Yes. Beefy prey. - All terrain. For eating beef. - Yes. - That if there was no beef in the early Cretaceous. - No. - We're so sorry. - No. - Dinosaur beef. - It would have probably enjoyed it. - Where was the beef? - This would make it one of the first Cretaceous hypercarnivorous ichthyosaurs. - Cool.

- So one of the earliest of the ichthyosaurs that started, 'cause this is not the only one known. - Right. - There were other ichthyosaurs with big, scary teeth for taking down larger prey. - There was a news not too long ago about an ichthyosaur caught in the act, eating something surprisingly large. - Yes. But this would be one of the earlier ones. I believe the earliest that we currently know of, in fact. - In the Cretaceous?

- In the Cretaceous. They were also able to note a number of differences from the type species of Platypterygias, which is... - The original species, the first one named in this genus. - Exactly. Platypterygias platydactylus. A bunch of these had to do with the stentation and the feeding strategies that it indicates, but a number of characteristics were able to remove this Columbian specimen from the genus. - So it's not Platypterygias? - No. They gave it a new genus, which is Chihitsuka.

So this is now Chihitsuka saccharum. - Oh, cool. So it's so different that they put it in its own genus. - Yes. - It also prompted them to present some new definitions for the overall group, the ophthalmosaurids, to keep that group defined well. - Right. Well, this is kind of like what we were just talking about with the ankylosaurids, where it was, yeah, ankylosaurids have hip bones that look like this. All right. Well, we just found one that has one that looks different.

So that no longer is a true... We have to expand what counts as an ankylosaur so we don't get confused in the future. This is, all right, we found an ophthalmosaurid ichthyosaur that's different. So now we have to expand our understanding of the diversity of ophthalmosaurid ichthyosaurs. - Exactly. So they propose some new definitions to guide that refinement of the group. The article did not get into it and I did not read far enough in the paper to get into it.

So... - Probably a lot of technical details. - Which is why... - As it usually is. - Didn't go looking for it. But as always, we will link to the press release in the blog post, so if you wanna go digging deeper you can. - Absolutely. They said this also gives some new ideas to how the group evolved overall. This ichthyosaur is aged to around the time when there's a major transition during the early Cretaceous, where we see a lot of the classic marine reptiles and marine big predator groups go away, like the short-necked plesiosaurs and marine crocs and a lot of the other ichthyosaurs to be replaced or to be taken the place of in those niches that they left open with the long-necked plesiosaurs and sea turtles and mosasaurs, but also this ichthyosaur, this hypercarnivorous ichthyosaur and others like it.

So it slots into that time and kind of adjusts some of the info for that transition. - Very cool. Now, the ophthalmosaurid ichthyosaurs are named for their eyes. They had very

large eyes. And as we've discussed previously, we discussed this in episode 116, that at least some of them, and I don't know about this one, but some of them are thought to have been divers. So this ichthyosaur potentially was a visitor to the ancient deep sea.

- It very well could have been. - Do you have any more to say about that? - You know, I think I might. Give me a little bit of time, give me a short break with a musical interlude, and I think I might have something. - And do you think you could talk for about an hour about it? - Yeah, yeah. I think I can pull that off. - All right, roll the music and we'll be right back.

- The deep sea, this falls under a group of those terms that I kind of love when we go over them, where like the term deep sea is a very straightforward. Deep sea. - Right. You go in the sea, the sea and you keep going down. - Yeah. - Deep. - Right. - I love it 'cause we know what that means, but what does it actually mean? - Right. Like the upper sky. - Yes, exactly. And so there's a special place in my heart for these terms when we go over them.

As a term, we all know what it means, but what does it actually mean? - Right. How deep is deep? - Yes, exactly. And it depends on what source you look at. Most I found, the deep sea is once you go below 200 meters. Some had it as a thousand, but most of the sources I found listed the deep sea counting everything below 200 meters. - 200 meters below sea level. - Below sea level. Starting at the top of the water, going down 200 meters or 600 feet or 650 feet.

And that's when the deep sea starts. - That's interesting because I am not an ocean expert, but I know that the sea floor has a habit of being kilometers down, so it would sound like that includes most of the sea. - Yes, it does. And this is true even if we push that limit, that definer to the thousand meter mark. It's still most of the sea below that. - Right. - And you are right that at least the number I found, and I'm sure that the percentage might be able to be rounded slightly one way or the other, but that this accounts for about 95% of Earth's living space.

Because it is most of the ocean and the ocean's most of the planet. - Right. Well, and the ocean is also a three dimensional habitat. - Yes. - Which is not quite the case in the same way for like a grassland. - Yeah. For most other places, you don't have this just, you can't operate in it like astronauts operate around the space station. - The full volume of the ocean is habitat. - Yes. But you may be wondering why the arbitrary number.

- I am wondering. - That's because the ocean is broken up into zones. - Are we going to, like Mr. Ray sings, let's name the zones of the open sea. - Absolutely. - There's, epipologic, epipologic, mesopologic, pathial, epistlepologic, all the rest are too deep for you and me to see. That's everything I know about the zones of the ocean. - All right. So now we can move on. - Comes from that song, Finding Nemo. And those are indeed the upper layers and they are sectioned out at meter distinctions and the top layer, the epipologic, the surface layer goes down 200 meters.

- Okay. - So for most definitions of the deep sea, it is everything that's not the surface. - Gotcha. - And this area is unique for receiving sunlight. It's sometimes called like the photic zone because it's where we get photons. - Yep. It's where the light comes. - This is where photosynthesis is happening in the ocean. And it is the part of the ocean that ranges temperature wise, like the surface changes temperature wise. - Oh yeah.

Cause it's going to be affected by the sun and the weather and the climate much more

heavily than things deeper down. - Yeah. It ranges from negative two degrees Celsius at the poles to 36 degrees Celsius around the equator. - Right. - So when you get to big ranges in temperature, that becomes basically not true as soon as you go below this. But we're done talking about that zone for the rest of the episode. - That's another episode.

- And goodbye. Mesopelagic is the layer beneath that. It goes from 200 to a thousand meters, 650 feet to 3,300 feet. And this is the area that we start seeing the loss of light. Though basically this is the zone we lose light. The top of it lights significantly, I mean, it's dimming all the way through the surface as well, but this area light is dimming all the way through. And by the time you reach the bottom of it, no more light.

- Gotcha. - This is the twilight zone or mid-water zone. So this is, when you hear people talk about the twilight zone in the ocean, this is the layer they're talking about. - Yeah. And I've heard that term too. I lied that not everything I knew about the ocean came from Mr. Ray. - Yeah. There's a lot of terms that pop up and this is where you start seeing bioluminescence come in. We'll talk about that later.

This is the bulk of where a lot of life is found. About 90% of fish will be found living in this area or operating this area. If not at all times, at times you will find 90% of the fish we know of swimming around this layer. So a very lively layer of the ocean. Beneath that. - Is it the bathial? - It is the bathial. - Yeah. That's one of my favorite things about Finding Nemo is that it was correct.

- Yes. No, it is. This is the bathypelagic layer zone. It goes from 1,000 meters to 4,000 meters down. - Okay. - So getting real deep. - Now we're measuring in kilometers. - Yes. This is from 3,300 to 13,100 feet down into the water. This is the aphotic zone. From here down, no more light. - No light. - We are in perpetual true darkness. This is often known as the midnight zone. - Okay. - Which means from this point on, the only light you get is bioluminescence.

- Is organisms making their own light? - Yes. It is also pretty much a constant temperature at this point in the ocean. We roughly have around four degrees Celsius, 39 degrees Fahrenheit. - Okay. - Very cold. - Almost freezing. - Almost freezing. - Obviously not quite. - Yep. - Sometimes we will dip below that in the deep sea, but it'll never freeze because of sea water. - Because of salt and stuff. - Physics of salt. - Right.

- We're also getting into some of the extreme pressures at this depth. - Oh yeah. - In this layer, we can be experiencing over a hundred times surface level air pressure, which equates to almost 6,000 pounds of pressure per square inch. - This is the amount of pressure that makes Aquaman so strong. - Yes, exactly. This is why he's so tough. - Right. - That's why you can't shoot this part of the ocean. - That's right.

This part of the ocean is bulletproof. That's why you can't shoot these fish. - Yes. - That's why giant, giant squids can't be taken down by conventional weapons. - Another thing that's really characteristic of this depth of the sea is that food becomes very scarce. - Yep. Because most of the food on the planet is living stuff and most of that sticks up where the sun is. - Yes. - Most of the food relies on sun as its energy source and we are now about as far away as we can get from that minus the bottom, which we're getting to, and we have not yet gotten to any alternate sources.

- Right. Oh yeah. We're in the middle. - Yes. So what we find here is marine snow that just flakes of food falling from the upper surface, food and poo. - Yeah, just organic matter that sprinkles down perpetually through the water column. - And that's the main

source of food at this level. So very little food, unsurprisingly life is extremely specialized at this level. You start seeing a lot of those viper fish and angler fish and those characteristic cliche deep sea fish are hanging out in this midnight zone.

- Angler fish. - And then we get to the abyssopelagic, the abyss. This is the abyssal zone, which reaches from 4,000 meters down to roughly 6,000 meters or AKA the sea floor, 13,100 feet to 19,700 feet. - Several kilometers slash miles down into the ocean floor. - This is the bottom layer of the ocean, generally speaking, we have one more zone. - Usually. - Pressures are in the extremes, we're over 600 times surface pressures. The water temperature is basically universally near freezing, like one in two degrees Celsius.

This is where you also get a lot of different specializations in life because now you are on the floor. We have a surface. - Right. - For the first time, we have things that you can sit upon. - There's a substrate. - So you get lots of things like octopus and sea cucumbers crawling around. You also get the tripod fish, which has special fins. - Oh yeah, for walking. - To prop, not walk, just prop itself.

- Oh, right, right, right. - To just rest on the sea floor and then swim off again and then just prop up again. - Just to hang out, just some kickstands. - Just to be weird. We will talk more about this zone. This is where most of the deep sea habitats, as you'd expect, that get discussed are found because now we can have different structures and different features. And then finally, and most extreme. - The megalodon zone.

- Yes, the hadal pelagic zone. - So hadal, H-A-D-A-L, as in Hades. - Yes. - Right. - This is the deepest of deep on our planet. This is as far down as things go, as far as we know. This goes from 6,000 meters, the average rough depth of the sea floor, give or take, down to 10,994 meters, roughly, or 19,700 feet to 36,070 feet-ish, which is the very bottom of the Marianas Trench. - The deepest place we know that you don't have to dig to get to.

- Exactly. This includes deep sea trenches, all the places that significantly dip below the sea floor depth that just go down for ridiculous extra distances on our surface. - For geology fans, refer back to episode 122, where we talked about why you get trenches, why the ocean floor dips. And for people who really liked our Aquaman reference, this is where the trench lived. - Yeah. This is where things get scary. So real quick, let's talk about the Marianas Trench, or Mariana Trench, depending on how you pronounce it.

This is a deep sea trench in the Pacific Ocean, the western Pacific. It's roughly crescent-shaped when you look at it on the map, and it runs a length of about 2,500 kilometers and 1,500 miles with some change. And at its widest point is about 69 kilometers wide or 43 miles wide. So it's big in and of itself, and at its maximum depth, a valley in the trench known as Challenger Deep, which is, if any of you've heard that word, this is what that word means.

It's not a separate trench. It's a specific spot in this trench. - It's the deepest trench in the deepest trench. - Exactly. This is the point that goes down to 36,000 feet. That's two kilometers deeper than Mount Everest is tall. - I was just about to say that. I was just about to say that. That's way deeper than Everest is tall above sea level. - That's one of those go-to vastly overused measurement comparisons that we often make fun of, like, how big is it compared to a bus?

But man, is that one so good. - Because Mount Everest, I want to say, is about 29,000 feet tall. - And so, yeah, this is a trench. You often hear about seamounts, which we'll

mention a little bit later, but seamounts are undersea mountains, some of which are much taller than our tall mountains on Earth. But this is a hole that's bigger than our tall mountain on Earth. That's so crazy. And it is the place, like I said, episode 122, where one giant section of ocean crust is dipping and sinking down underneath another giant section of ocean crust.

And that dip is what causes that trench to exist, that low point in the topography. - Now, something we will mention is that studying deep sea life is difficult in and of itself. But we actually know a good bit about a lot of deep sea life. We have taken expeditions down there and brought stuff back up. When it comes to trench life, we know very little. It is especially hard to get down there, and life's very rare at this depth.

There's not a lot to be found. We do find single-celled organisms, unsurprisingly. Cells can survive basically anywhere so far that we've found. So they are down there, but there's a very short list of vertebrates, of fish and other bony animals that have been discovered at this depth. Most of the seafloor at these deep sea levels are just going to be covered in mud. Not sandy seafloor, but mud or ooze, as it's sometimes called, which is mud with lots of organic material in it.

So it's kind of slimy. We don't have many sandy environments in the deep sea. Sand's too heavy for currents to carry this far into the ocean. So if it's not muddy or oozey, it's typically going to be rocky. Places where there is no substrate. These are either on steep surfaces, the edges of islands and continents, seamounts and valleys and things like that where sediment doesn't want to settle because it's not flat enough. The areas you find that are flat, that are rocky, are typically mid-ocean ridges and rift valleys where the rock's too new to have developed a muddy surface.

Right. That is young rock. Yes. That hasn't developed a layer yet. So now that we are on the seafloor, let's talk about some of the habitats that you tend to find life occupying here. I'm excited to talk about this because the more you talked about describing the structure of the deep ocean, the more I'm realizing, and I hadn't really thought about this, how much this discussion has in common with our discussion about caves in episode 112.

That you're getting down to places that are far enough from the surface that the temperature becomes consistent and you don't have light and you don't have resources and you don't get a lot of sediment. It's an interesting parallel that I hadn't really connected in my head. Yeah. Absolutely. There are a lot of similarities. It's an extreme alien bizarre environment with extreme weird alien creatures. Now most of the seafloor is just seafloor. What is often called the abyssal plane because it's D&D levels of awesome here.

The seafloor is on average between depths of three and 6,000 meters. So 9,000 and 19,000 feet. It's going up and down, it's having bumps and hills and valleys, but most of it is a pretty much flat, fairly featureless, just stretch of ocean floor. I've often heard this compared to a desert in the fact that not much food is to be found because you don't find anything growing here. The food that you do have access to is that marine snow.

What small percentage of it makes it all the way to the bottom and doesn't get eaten by filter feeders above you? It's estimated that less than 5% of the food produced at the surface actually makes it down to this depth. So you are getting the scraps of the scraps that actually make it all the way down through 6,000 meters of water. You do find life here. You find a lot of scurrying around, or I say scurrying, sea cucumbers don't

really scurry, but crawling around.

Laying around. Yes. And so you find lots of seafloor life like cucumbers, and I know brittle stars are very common at the deep sea, crustaceans, like squat lobsters and prawns and stuff like that. When I was looking for images for the teaser for this episode for our social media, most of the images that you will find are pictures of a seafloor, but where there's light. Yeah, where you can like dive down to it. Where you see coral and sponges.

And so the typical image of a seafloor that you might have from documentaries and stuff is not what we're talking about here because we are too deep. That was the issue I had. I was looking around, I was like, "Why can't I find any pictures of the deep seafloor?" And then I went, "Oh, because there's no light down there." And there, you can't take a picture down there. So the abyssal plane is a mostly featureless, mostly lifeless, mostly foodless.

God, barren. It is barren just about. Yet pockmarked around the seafloor, there are some unique enough or momentary habitats, times where life has a reason to gather or resources are concentrated that allows life to gather in denser populations. Little oases in the desert. One of the most simple of these is seamounts, areas of the seafloor that just go up a bunch. Undersea mountains. So they are just tall peaks. Sometimes this could be the edge of an island that actually breaks to the surface.

But the fact that the seafloor suddenly goes whoop means that ocean currents hit it and go whoop, and it will cause updrafts of currents and concentrations of currents. So any nutrition in the water gets concentrated against the edge of the seamount. And there are whole populations, especially in the upper reaches of the deep sea, that congregate around seamounts and are found almost exclusively around seamounts because of this concentration of ocean resources. You get an upwelling of nutrients.

Exactly. It's like an updraft in the air. Yep. But it's carrying all the stuff that's floating around the water. All the yummy stuff. And it's concentrated. Another thing, another environment that you find in the deep sea, in comparison to what you were mentioning about most seafloor things you see are things with like coral and stuff. There are deep sea reefs. There are. I learned that not too long ago. They are not the same as our reefs.

These are constructed by what are called deep sea corals or sometimes cold water corals, because most coral reefs are warm waters and shallow waters. They need the depth so that they have sunlight and they tend to be very particular to temperature. So they want warm, not necessarily tropical all the time waters, but warmer than cold usually. Episode 36. These corals can be found down to depths of 6,000 feet, at the very bottom depths, not just the edges of the deep sea.

Deep coral. And they can be found in all the varieties that you find coral in the shallows as singular polyps, kind of like a sea anemone, as a colony or as a reef of multiple colonies. They often form associations with sponges to create structures growing alongside them. The main thing that's weird about them is they don't use sunlight. They are not photosynthesizing. They are purely eating. So they are only catching food with their polyps, which normal corals up the surface do as well.

Right, filter feeding. Yeah, at night they catch food with their little stinging polyps and bring them into little bellies. And then in the daytime they pull the polyps in and they photosynthesize with algae that lives in their body. These corals have no algae because they have no sunlight. So they are just eating all the time and that's it. Probably

scavenging or carnivorous, depending on what they're catching. I was going to call them, first I was going to call them cold coral.

Yes. And then you said that and I wanted to call them carnivorous coral, but they're catching like little drops and snow and horses. They're probably mostly eating fish poop. Right. Not quite predatory coral, but if you're a Hollywood producer, let us know. We'll come up with an idea for a coral horror movie. Yes. And like reefs in the shallows, these deep sea reefs also provide homes and habitats for many forms of life. Yeah. Their shelter, their substrate, their collections of organic matter, I would imagine.

So you'll find a denser collection of life around these. By the way, for anyone listening who might not know, when I say the word substrate, substrate refers to a surface on which an organism lives. Yes, yes, yes. So that could be sand, it could be rock, it could be coral, it could be bone. You know, whatever an organism settles upon, that is the substrate. Yes. So coral can be a substrate for other things to live upon.

Absolutely. There are specialists that grow on or within coral colony structures. The rest of the examples of deep sea habitats are typically when an influx of nutrition, of resources are brought to a specific spot of the deep sea. And this happens a number of different ways. Some come from above, some come from below. And the ones that come from above include things that fall down in their hole, in their entirety. Right. Not just in little microscopic fragments.

Yes. But when a body or other organic structure hits the bottom of the ocean, these are called food falls. Yes. They include anything that falls down there that is organic. So it could be a fish, it could be a log. Yep. It happens to come down. But the most common you'll hear discussed are whale falls. Yeah. Because when a fish falls down, that's a lot of food, that's a much bigger portion of food than life typically gets.

But when a whale falls down, that's going to change that area of the sea floor for life for years. A whale carcass dropping to the sea floor creates a long-term habitat. Yes. Whale falls are a huge subject of study because they are one of the most dense collections of deep sea life at any point, basically. And they go through a life cycle. When the whale lands, different things are eating on it than will be eating on it in a few years when it's been chewed away mostly.

And they've actually broken it down into three sub-stages. The first is called the mobile scavenger stage. Yep. Sharks and fish and stuff that moves and swims and comes and takes bites. These are things that smell or sense the whale fall and swim to it or crawl to it and then start chewing on it with mouths. Sharks, hagfish are extremely common at whale falls. These are those boneless, jawless fish that latch on and just pull chunks off with a raspy tongue.

They have tons of crustaceans and other invertebrates and arthropods, deep sea isopods and stuff, which are the cousins of pill bugs and roly polys, but are like the size of not a long loaf of bread, but one of those stubby loaves of bread. These come in from all around. It can sometimes be just hours after landing that they start showing up. Like a signal has gone out and just deep sea organisms come together to what can be, I mean a whale, it can be 20, 30, 40, 80 feet of animal that that's a huge amount of food.

Yeah. Massive, massive influx of organic. And just like out the surface, it'll draw in scavengers from all around. These scavengers can remove up to 40 to 60 kilograms of whale a day. Wow. Just chowing down. This process typically lasts for months, potentially years depending on who gets there, but the first section of the whale's, the

whale falls life, second life is being chewed on by these scavengers. Of the whale's death. The next stage is the enrichment opportunist stage.

Once again, during a duration of months to years after or during the scavengers chewing pieces and pulling chunks off and eating the whale, the surrounding area, the sediment around the whale becomes enriched with all the bits and juices and goo from the whale. So you have this soiled spot of mud that's just full of whaley goodness. And so you will get organisms that populate and colonize this enriched area. Meat worms are very common. Other crustaceans that can settle down are very common and will start growing off all of the bits left over.

Now it is a very nutritious substrate. Yes. This also includes things that will start growing on the bones to get the last little bits of stuff there. The worms that populate the sand have been counted up with a number of up to 40,000 in a square meter, which is one of the densest assemblages of life in the world. Like it's like the Tokyo of ocean habitats. Yes. So just life stacked on top of itself. And then we hit the final stage, the sulfophilic stage.

The best stage. This is where it gets weird. This stage can last for decades. Since all the juicy bits have been taken away, all the meat, all the cartilage, all the fat, all the skin has been eaten by the larger scavengers and more mobile scavengers. Typically all you have left is bones and you may be thinking to yourself, Hey, that's how we get a whale fossil right there because now there's nothing left to eat. Right? Well, you'd think so.

Nope. Because there's anti bone forces out there. Yes. And then lipids are full of nutrition and there are sulfides, sulfur molecules that come off of bones that can be drawn from bones. And there are bacteria, chemo autotrophic bacteria that specialize in using sulfides to derive energy. This is going to be a theme throughout this episode. We're going to talk a lot about sulfur sulfophilic chemo autotrophic bacteria. Yep. If there's sulfur, I can eat it and grow upon the bone.

This is the life that is producing energy without photosynthesis. We're doing it with chemicals, not the sun, and they are able to take sulfide, make sugars and foods. And some of these bacterias you'll find growing just by themselves, but you'll also find hosts, symbiotic relationships with invertebrates that contain or host a colony of this bacteria as they grow on or into the bone to then let the bacteria eat the bone. And they benefit from some of the extra nutrition that they produce.

The most commonly discussed of these are the bone worms. Osidax. Osidax worms. That's not like a pop culture reference that I just made. Osidax. That's the name of the worms. That's the name. Osidax worms. One of the coolest names of an organism anywhere. These are small, itty-bitty worms, like very thin worms. As larvae, they just ride the ocean currents and settle upon exposed bone. The first that land become females. They then tunnel into the bone with root-like structures at one end of their body, filaments that bury themselves into the bone.

And then the other end has a feather-like structure that is collecting oxygen from the water. So it looks like a little sprout, a little stem with like little leafy things, but those are their gills. And then root-like worm growths into the bone. Those roots carry the bacteria into the bone to expose more area and let them eat the bone, and the worm benefits from the nutrition. The next worms to arrive become males and land on the females and then grow inside the females as parasites and provide them the ability to mate.

The communities found on whale falls, these worms and the ones that can leave, are very interesting because no two whale falls are ever the same. Because it's not something that you don't fall, the whale doesn't fall and then the animals already there go onto it. It is a collection of who gets there. Whatever happens to arrive in time. And we're learning more every day. Every time we discover a new one, new species are discovered. New bone worms are often discovered because it's far and few between that we observe a whale fall, even though estimates suggest that they actually are fairly common in the ocean.

Probably a whale fall every five to 16 kilometers in the Pacific. Wow. I mean, there are a lot of whales. There are a lot of whales. And they have, I found a couple of, a number of sources that mentioned that the number of whale falls was probably significantly affected when we started whaling. I was just wondering that. Probably not as many as there used to be. And we don't know what effect that's had on things because we didn't know about whale falls before that.

Oops. But though we find them irregularly, they are probably more common than we are seeing them. But it's like the needle in the haystack situation. We can't, we can't aim for whale falls. They don't happen in a specific spot. Right. Wherever it died, which could be anywhere there's ocean water, we just have to hope we happen upon it with a sub. Now this is very similar for other animal falls, but we also have wood falls. And this may be my new favorite deep sea community.

When logs make it out into the ocean, which doesn't happen as often as whales do, there aren't as many sea going trees. Not very many. Maybe they're there by mistake. But when storms wash material from land to sea, when a tree makes its way into a river and is carried out to sea, we will find ocean logs. And once they become saturated enough to stop floating and they go all the way down, they will also develop their own special communities.

And these communities are a little bit different because it's wood. It is a bit tougher than flesh. And a lot of these are more entrenched. You don't get new sharks coming along. There's not many swimming wood scavengers. But one of the big reasons it's weird to me is a lot of the organisms found on deep sea logs are only known from deep sea logs. About 90% of them seem to be wood specialists, which means this is an ecosystem in the deep sea specialized for eating food that only comes from the land.

And has to have a big chunk of that sunlight stored energy to be thrown out to the sea. Which is super weird. Which implies, of course, that this is something common enough that there are organisms that have evolved to adapt to it. Precisely. You also see some sulfate, sulfide bacterias and bacteria symbionic organisms. But the most common member of the wood community are xylophaga bivalves. Bivalves include all your two shelled animals. Slams, oysters, mussels, et cetera.

These have a specialized shell that doesn't meet up. They're curved. And the edge has denticles. Has teeth on it. And they use the shell to chisel away wood with their muscular foot. They chisel away wood with the shell and whatever splinters or pulp comes off of it gets collected in their stomach-like structure, which is full of wood eating bacteria. And that's what they digest. The bacteria lives in their gills and digests the wood. Wood scraping clams.

There's also a bunch of snails that are known from there and sea stars known from them. And these communities seem to be much slower to develop. Like it could take a

couple of years for a log to develop a significant deep sea community. So it's not like a whale where everyone goes a log and rushes over. Because these can't rush. Most of them are landing there as larva. And I found one article where they were quoting some of the researchers that dropped logs down in spots to then come back and collect them.

When the question was brought up, what are they doing when there isn't wood? And they said, we don't know. We don't know. They're hiding somewhere. They are not eating wood. And that's the case with a lot of these communities. What's happening in between when they're living on one of these? I don't know. Because that's not where we see them. Yep. Nope. Because it is a lot of ocean out there for them to be doing other stuff.

The other examples of massive nutrition coming into the deep sea are when it's coming from beneath ground. When things from the earth. When the creatures from deep within the earth come up. No, that's not what happens. When they rise for their vengeance, the deep ones. This is when minerals, usable minerals and usable chemicals are released from earth's surface in a variety of ways. And the surface meaning in this case, the floor. Yes. The sea floor. The crust.

There are three main types of these. The most common you'll hear about are hydrothermal vents. These are areas of the crust, typically around spreading zones and where the crust is thinnest. Where water can seep down through the rock and get close to magma and get super heated. And while it's being super heated, it will start to absorb minerals and metals from the rock and just become infused with zinc and iron and copper and lead and cobalt and all sorts of good yummy metals.

Then it's now hot water and hot water rises. So a geyser. Pumps its way out through the cracks in the rock. And as it hits the cold water, those metals will precipitate out of the water and be deposited and eventually will form structures around the water flow and form chimneys. So it's sort of forming like a volcano where as the lava comes out, it's cooling and creating a volcano. In this case, it's the minerals coming out and creating, like you said, a chimney or a mound built of those yummy things.

Yes. These are typically separated into black smokers, which are pumping out iron sulfide and are therefore black in color, and white smokers, which are mostly barium and calcium and silicon and are more white in color. Black smokers are what mostly I see discussed and shown pictures of. These are fairly recently known structures. We only discovered these in 1977. So we are still learning a ton about these structures and how they function. The water being pumped out of these chimneys is extremely hot, 400 degrees Celsius or 752 degrees Fahrenheit, which I saw listed as hot enough to melt lead.

Oh, very hot enough to melt sea creatures. Yes. And there is sea life surrounding these chimneys. Most of them relying on that sulfide, that sulfophilic bacteria or a version of sulfophilic bacteria, though there are others that are surviving on other minerals on other chimneys, chemoautotrophic bacteria. Most of the life, though, is not found at the 400 degree Celsius temperature. Right. They're found from 8 to 25 degrees Celsius, which is still warm. Pompeii worms, which are the worms that grow alongside the tubes in one specific group, have been found up to 60 degrees Celsius.

And then there have been bacteria found up in areas that get up to 120 degrees Celsius. Of course there have. Yes. Bacteria. Yes. Bacteria like those extreme habitats, all sorts

of life are found here, all hosting or living off of these various bacterias, clams, mussels, shrimps, those big tube worms that grow kind of like a stock. Yeah. And some of them build their own tubes to live in. Absolutely. And then you also get other bigger animals, eels and fish and crustaceans that come in to then feed off of the other animals or graze.

Because yeah, like you said earlier, the sea floor is so rarely nutritious. This is a space where a non sunlight source of energy allows for an entire food chain where you have those bacteria and then you have things that benefit from the bacteria and then you have grazers and predators and all sorts of different and decomposers, I'm sure. Yes. Where you can have a full community and food web because you have a rich source of nutrients.

Absolutely. And this is not the only source of this kind of nutrients. You also get cold seeps, very similar concepts. This is areas where various things in the crust are seeping up to the surface, typically fluids or gases. This could be methane or petroleum based chemicals, molecules coming up. They form around cracks in the crust. The reason they're called cold seeps is not because they are super cold. It's because they're colder than the vents. Right. Because everything down there is already extremely cold.

Yes. Almost frozen. Yep. So it can't get a whole lot colder than that. So the vents are when things are being pumped up because magma is interacting with water. This is where things are just oozing up, but it's just at a more normal deep sea temperature. You find microbes and lots of mussels and clams, but also their own tube worms growing around these seeps, once again, with their own chemo autotrophic bacteria, handling the methane or sulfides that you find there.

And then the final of these seepy-uppy nutrition sites are brine lakes, which is probably my second favorite of all of the deep sea habitats now. I think I remember this from a documentary. Brine lakes are called lakes because they are a section of super salty water, just bitter about everything. This is sections of water that are hyper saline, hyper salty, so much so that it will not mix with normal ocean water. Yeah. This is the documentary footage that I was thinking of where you can see a fish swim down into a lake.

Yes. At the bottom of the ocean. It goes, and you can see the boundary between the ocean water and the lake water underneath it. And there's a surface. It is notable. Now this is the reason that I think they're so cool. This is not super salty water that sunk to the bottom of the ocean. This is areas where salt seeped up into the ocean. So one of the well-known brine lakes is in the Gulf of Mexico and in the Jurassic, this area was a shallow sea.

At some point that sea got cut off. And when this happens, we get salt plains. The sea eventually evaporates and you have a wide open area that was once sea floor now covered in all the salt that was in that sea, sometimes in extremely thick layers. Over time, this salt plain, this ancient sea was covered by younger sediment and buried. So by the time the sea made it back into the Gulf, it was separated by a layer of earth.

Well, more recently when the Rocky Mountains began to rise up after that and erode that influx of sediment into the Gulf from the Mississippi, added enough weight to put enough pressure to break the seal and cause salt to just ooze up, to seep up through the floor into the sea water, into the ocean water. And when it mixed, it became hyper salty and did not continue to mix with the water around it. So brine lakes are areas of

ancient seas that are seeping up through the floor and becoming under deep sea lakes.

It sounds like it forms in a somewhat similar way to what we discussed in episode 67 with the La Brea Tar Pits. When you have that asphalt seeping up and forming these pools of this tar-like substance. In this case, it's pools of hyper salty water. And much like the Tar Pits, they are incredibly deadly. Yeah, I would imagine. There is no known animal life that we're aware of that can actually live in the brine lakes. If I remember that documentary, I don't remember what documentary this was.

It was probably like Blue Planet or something like that. If I remember correctly, something died in the lake in that documentary segment. If I remember, because I think I know a scene, an eel swims too close and then goes into chemical shock because it's so salty, it's deadly to all animal life we know of. And then once they die, they are pickled in the brine lake. But along the edges of the lake, there's tons of life.

These lakes are a high source of methane. Once again, we come back to methane and there are bacteria that like methane as we've mentioned. And so there are tons of, lots of mussels and shellfish that contain methane and will just grow in huge clusters around the edge of the lakes. So they will encircle the brine lake, but nothing goes into the brine lake. And we do see actually similar communities, similar organisms in these various, the seeps, the vents in the brine lakes will often have overlap in the animals and organisms we see occupying them.

So those deep sea habitats are all unique and extreme in their own ways. But so is just the deep sea. It is a very extreme habitat and biome of earth for life to be living in. And because of this, the animals and organisms that live down there have developed some particular adaptations and features to survive in the deep sea. So I just wanted to go through some of those real quick. We won't be able to go through all the interesting life that is in the deep sea because there's too much of it.

And we've already been talking for a little while. That material for future episodes. Absolutely. One of the first obvious ones is light. Bioluminescence is incredibly common in the deep sea, as we already mentioned. It's seen being used for a number of different purposes, from finding food to avoiding food to finding mates and communicating. It's found all over the bodies of the animals. You have lures, those little specialized dorsal fins that come out like a little fishing rod and glow at the end.

And the angler fish style. Counter illumination, which is one of my favorites, where a organism uses its light to break up its outline. The firefly squid has lots of little dots that breaks up the shape of a squid when it lights up. But the hatchet fish has lights on its belly to mimic the bare amount of sunlight that comes down to the depth in the twilight zone and makes them blend in to surface light when viewed from below.

You can have confusion tactics, the ones that will shoot out decoy blobs of light to try to juke the predator. One that I saw called burglar alarms, where they will spray predators with bioluminescent material so that they light up. Oh yeah. Well, that's cool. So that they're more easily identified. And the reason it's called burglar alarms is that the hope potentially is that the police, a bigger predator, will come in and take that predator. And then you have headlights where they have lights around the eyes.

These are really common among lantern fish and stuff like that. Dragon fish have it and dragon fish is my favorite because one of the other things you get from having low to no light in the deep sea is that blue light is really all you have left. Right. Because that's the wavelength of light that goes farthest in water. Something else gets absorbed. So there

is no red, there's no green, it's just blue. Which means a lot of organisms down there are red because they're going to appear black in the blue light.

Yeah. They're not, you can't see the red until you bring them up in one of our lights. Right. Into full light. Yes. So down at the deep, they just look black. And most animals have lost the ability to perceive red light because you don't need to because it's not there, except the dragon fish, which still can see red light and produces red bioluminescence. So it has spotlights that no one else can see it shining. Oh, that's pretty cool.

So it can search food out and they don't realize they've been spotted because they can't see it's light, but it can. Wow. It's like having night vision goggles. Exactly. Yeah. This is the equivalent of night vision goggles when there is no light. The other thing that life has to deal with down there is the pressure. The extreme pressure does have an effect on biology. One thing is that it'll compress any air pockets. So like swim bladders that fish use to regulate their buoyancy would get compressed significantly, which is why a lot of deep sea fish don't have air pockets in their body.

They don't have a swim bladder, which is not, you know, there are other fish who have lost swim bladders, but that is a inconvenience down that deep because air pressure is not the same anymore. You also have issues with biomolecules. Biomolecules in the body can get deformed under such extreme pressures and stop functioning correctly. So a lot of deep sea fish have special chemicals or molecules that they produce in their body that help fight these pressure conditions.

One of the common ones is often abbreviated as TMAO, which is the thing that makes fish smell like fish. That's the chemical that does that. And we find much higher degrees of it on average in deep sea fish versus shallow fish. So they're smellier fish. They would smell more like fish than other fish, but it works two ways. Those chemicals are really good at deep waters and high pressures, not very good in low pressure. So this does kind of lock those molecules to functioning at those pressures.

We see a similar thing with temperature because you're dealing with very cold temperatures. And we mentioned about adaptations to cold temperatures when we talked about the poles. - That's right. Episode 114. - You see a lot of similar adaptations here. So if you listen to that episode, apply those to a lot of deep sea organisms and it's very true. - Anti-freezing your blood and all that kind of stuff. - One of the things that you get here is that the various membranes in the body, you don't want to get stiff in the cold water because then they won't be functioning correctly.

You want them to be semi-permeable. You want them to be flexible like membrane should be. So a lot of deep sea fish have, let's see, where is it, have unsaturated membranes instead of saturated membranes, unsaturated fat and saturated fat is what that's referring to. The comparison I saw made was that it's like butter versus olive oil. Butter gets hard when you put it in the freezer. Not quite frozen usually, you could still break it apart. There's not ice crystals there, but it is tough.

It wouldn't act well as a membrane anymore 'cause it is no longer semi-fluid. While olive oil, even in the freezer, is still a bit more malleable and butter is a saturated fat, olive oil is an unsaturated fat. But just like those two, when brought up to room temperature, butter retains its shape and is good. Olive oil is a liquid. So when warmed up, these membranes would not function as well. - So these are animals that in some respects are stuck down at the depths.

- Yes, absolutely. One of the other limitations is the source of food. We already mentioned a lot of the ways by finding sources of food that animals deal with this. There are a lot of marine snow specialists for collecting food, but you also see lots of adaptations for saving energy. Lots of very slow swimming, lots of basically no swimming. They only swim when they need to and otherwise they just sit there. You also see that a lot of them have very low muscle density, which means it's less energy to maintain them.

They're weaker, but they get the job done. And then you see a lot that use what's called vertical migration. They go from the deep to the shallow in the night, feed on all that surface water food, and then go back down during the day to hide in the deep. - I think a lot of squid do that. - A lot of squid do that. - That's the animals I think of. - There's a lot of plankton that make that migration.

And they described that there's often a layer of foraminifera that try to catch the plankton as they make their migrations up and down. - Cool. - They call it a minefield. - Yeah. Ooh, mate. That's really cool. - There's also vertebrates, sperm whales, and beaked whales that do the opposite, that dive down to feed and then come back up to breathe. And so there's a lot of weird ways life has adapted and this is not all of them.

There are other interesting extreme features, but if they're topics for future episodes as you said, but life has gone through a lot to be able to live down at these depths, to be able to sustain themselves and continue to survive in such a distant and oddly isolated environment considering that it is most of the planet. So next, after a break, let's talk about that. How did life get to be the way it is? What are some of the evolutionary trends that we see with deep sea life?

And what can we tell about ancient deep sea life from fossils? - Deep sea and deep time. - Together finally. Unsurprisingly, our understanding of deep sea fossil history and therefore evolution is a bit patchy. - What? - Yeah, the fossil record for the deep sea ain't great. Some would even call it poor. Yeah, we don't have a huge diverse and robust fossil record for the deep sea. It's the same thing as open sea issues that a lot of times even if you would fossil there, we may never get to a point where we get to see it because you're still underwater.

So the fossils that come from there are far and few between. We do have deep sea fossils, things that are thought to have been fossilized at depths that would count as deep sea. But this is also hampered by the fact that today the deep sea is difficult to study. Modern day, right now deep sea is not easy to reach nor to spend large amounts of time to research. For instance, when it comes to the Hadal zone, the trenches, we still don't fully understand how life survives at that level of pressure and lack of food.

There are areas down there that get very little to no oxygen because it is so far from the rest of the ocean and there's basically no currents that we have found life that seems to function without any oxygen and we're not sure what they are functioning off of. - Life's so extreme that it's very mysterious. - Yes. It's like asking what are aliens like? I don't know. Well, what are those animals like? I don't know. And part of this is just because there's only ever been two human led trips down into the Marianas Trench, one more recently and famously in 2012 by James Cameron.

But the original one was by Jacques Piccard and Don Walsh in 1960 and they went down 11,000 meters and had to stop when the windows started to crack, which, hey, nightmare fuel. - Yeah, that's when it's time to turn back. - The rest of you can take a

break because that's it. So we've only ever been down there, like down, down there a couple of times, literally. So we just don't know a lot. So it's hard to study today and we don't have a lot to study from the past.

- It makes sense that deep sea fossils would be rare because that's not a very life rich environment and also it's way down in the deep sea. But of course, ocean floor rises in some cases over time with continental building and such, and you can get various ocean deposits in places where we can access them up on continents. We have tons of, you know, there's tons of places where you have ocean fossils up on continents because the land has uplifted over time.

And you can even study the organisms in those ancient sediments to estimate the paleo depth, the paleo bathymetry. So obviously if you find like coral reefs, you're in a photic zone. But a lot of this kind of study is done with things like foraminifera, where you can look at your community of foraminifera and based on what we know about their preferred habitats, get an estimate that yeah, this particular rock formation is not just ancient ocean, but a thousand meters deep ocean or 2000 meters deep ancient ocean or something like that.

So we can get the occasional deep sea environment in our fossil record. That's how we're able to identify whether a fossil is deep sea when we find it is looking at the foraminifera around it. And we're doing that by comparing it to today's foraminifera. So it's a kind of two pronged analysis. As mentioned in part one, there's been lots of research on the biological adaptations life has made to survive at those depths. So there's tons of research on the biochemistry and the physiology of deep sea life.

But when it comes to just general evolution down there, it goes back to what you're saying about caves, where there is not one trend because there is not the deep sea habitat. It is all of the deep sea, but in general, it is pretty much, or most of the evidence seems to agree that the deep sea we know today, the life down there today, mostly came about or started the, you know, the ancestral form of it formed in the late Mesozoic or early Cenozoic.

Oh, interesting. Which is a surprisingly recent formation for the deep sea community. Yeah, that makes it only about as old as modern, like the modern style of plant communities with flowering plants. Yes, precisely. And that's not saying there wasn't deep sea life before that, but the modern assemblage that we recognize, the groups we recognize down there and the habitats that we recognize formed the way they are. Right. And that does actually make a bit more sense to me intuitively because I remember reading about early Paleozoic ocean communities sort of gradually making their way into the deeper waters and the Mesozoic era saw the rise of a lot of modern groups of ocean organisms.

So it sort of intuitively makes sense that as modern groups took over the oceans, they did then eventually, maybe a little later, colonize the deep sea. Yes. And a hypothesis that came up around this idea and in conjunction with extinction events that would have made way for the modern assemblage to occupy the deep sea, is that there are a number of potential extinction events in the Jurassic and Cretaceous with the oceanic anoxic events where there was a significant lack of oxygen that could have majorly affected the deep sea and a period of cooling in the mid Cenozoic of deep sea water.

And these events, I think some of them are proposed events, I don't know how well supported each one is, but these events are thought to have been potential extinction

events for deep sea communities and that after each extinction event, the deep sea would be majorly hit with loss of species and then recolonized by shallow water species. Right. Things have to readapt into those habitats. It has to be filled in again, not by the deep sea diversifying back out, but by being filled in by new organisms from shallow waters.

This also went along with the idea that the deep sea is a place that has preserved ancient shallow water organisms, that if something from the mid Cenozoic came down from the shallows into the deep sea, they might have gone extinct in the shallows, but have remained in the depths. And I think it acts as a refugium for at least the later version of whatever it was, the shallow water crab moves into the deep sea, presumably it is evolving into a more adapted form for the deep sea.

And that might be all that remains eventually of this group of crabs. Yes, absolutely. And we do see some things that seem to kind of support this idea that there are ancient groups locked away in the deep sea, things that also support the idea that evolution down there is incredibly stable and things are just kind of locked away in time. One of the most poignant, I guess, or literal examples of that was there was an article that said, "A living fossil found in the deep sea."

And we've had a whole conversation about that term. The whole episode, episode 90. And we're not huge fans of it, but here there actually is, it's kind of blatant. There was a trace fossil or potential trace fossil from the Eocene that was a deep sea structure of some sort that consisted of a bunch of holes, potentially burrows, on the surface of the sediment. And we have the full thing, so we have it, we can observe it in 3D.

And then on the surface, though, the holes took a hexagonal six-sided pattern to it, rows of hexagons with each line of holes forming 120 degree angles with one another, tiny holes like one millimeter, and little shafts that go down just a couple of millimeters. And it was considered a trace fossil of burrows, probably not from an animal but probably bacteria that was forming these structures or something like that. Or maybe there was something burrowing and giving a place for bacteria to grow, to then graze on it.

Very odd structure. The pictures of it are very weird. Well, much more recently, a camera was able to capture a picture of an identical structure on a mid-Atlantic ridge over 3,000 meters down. Wow. Now we haven't gotten to examine that one because it's just a picture. Right. But it looks basically identical to this Eocene fossil. So whatever type of microbe it might be, there are microbes today living that same way. Yes. Or if it's a body fossil, because it could be that.

Right. Little do we understand this. It could be a body fossil of a sponge or a type of foraminifera called xenophyllophores, which create their cast, their shell from the sediment and could have formed that pattern. So we don't know what it is. It could be a trace fossil. It could be a sponge. It could be a foraminifera. But whatever it is, it seems to still be around today or something extremely similar doing a very similar thing is around today.

So there is some evidence of like extreme stability and ancient groups persisting in the deep. But then there's also lots of research that suggests, no, A, the deep has not been just that stagnant. And there seems to be a lot of groups that have done their diversifying in the deep without need to be recolonized. Brittle stars, which are the little thin armed cousins of sea stars. Very common and like of deep sea fossil organisms,

they're one of the most common because their whole body is good at fossilizing because it's hard bits.

And they're all over the deep sea. There was some fossils from the Jurassic from various countries gave some evidence that between shallow water and deep sea brittle stars, which is the cool part, seemed that evolution in that lineage, the early evolution took place in the deep that they had already diversified and split off while in the deep water. So not that they made it to the deep, but that they started there basically. They got there and then diversified in exploring this new habitat that they had moved into.

And that this brittle star fossil was one that evolved from other deep water brittle stars. We also find examples of communities from the deep sea. There was a cretaceous community fossil from the mid Atlantic that was a community of echinoderms, other brittle stars, as well as other echinoderms. So I didn't see a specific list of every kind. Once again, from this is about 1000 meters depth to 1500. And the fossils are about 114 million years old.

And they look basically the same as today's deep water echinoderms, which once again doesn't support the fact that they were going extinct and being replaced multiple times. That seems that no, that community's probably been pretty well situated, at least in this case since the cretaceous. This is also goes against the recolonization hypothesis because a lot of those groups, most of them don't have any fossil records in shallow waters after that point. So the fact that we have similar communities then in today, but no one from the shallows to recolonize in between then and today means that they've been there.

That is a deep sea lineage of life. Exactly. And so there's multiple examples like this that seem to go against the idea that the deep sea has been constantly recolonized and not been diversifying itself in the deep. But it sounds like that's also still something that is being debated and being discussed. It wouldn't surprise me if it's different for different groups. Absolutely. And even potentially different areas of the ocean, like or at different times, the pattern changes and like you said before, it's a lot of habitat and there's a lot of organisms there.

And some of them seem to be surprisingly stable, lasting for millions and millions of years. And then others, there is evidence of effects like global cooling having significant effects on the deep sea and potentially that it is not immune to changes in the earth like we typically thought. We used to think that it was so far away that the extinction is just aren't going to reach down there. Right. So if there is land ecosystems collapse, then it's really not going to affect them down there.

Yeah. But there is evidence that it actually is affected and maybe potentially significantly affected. So we may see similar, if not the same extinction patterns in the deep sea that we see in other fossil habitats. But these things are still being researched heavily and with a lack of evidence and accessibility, it makes it difficult. Now, though the fossils in the deep are rare, we do get fossilization if not just of singular organisms of the habitats from time to time.

One of the most common ones we have fossils of is whale falls, which are still very scarce. There's like a handful known from each age, so it's still not a lot, but we do get fossil whale fall deposits. Typically it is identified as a whale fall when we have a whale skeleton with something fossilized on it. Right. Evidence of those stages of whale fall community. And at least from all the examples I came across, they pretty much all seem

to be the second or third stage, because you're not likely to preserve the shark that's eating the meat while it's there.

You're preserving the things that are growing on the bones, the worms and all those things and the clams and the muscles that are attached to the skeleton. One of the interesting things with this is that we do see major changes in these over time. Food falls in general, because there were no whales for a long time. And we do have food fall fossils from other organisms, plesiosaurs and ichthyosaurs and ancient marine reptiles and stuff like that.

There was recent research in 2019 where they sunk the alligators. Mm-hmm. Yeah, we talked about that. We talked about that in the news, and that was to try to see what things might have been feeding on on those ancient reptiles and our alligators and crocodiles that make the way out the ocean still preserving those communities. Right. Is a reptile fall different from a whale fall? Yes. Sea turtles would also fall in that category. But one of the other interesting things is even among whales, we see a shift as whale evolution goes along.

You know, because the earliest big whales, like balsillosaurus, showed up 40 million years ago in the Eocene. And we have individual fossils, but I didn't find any listings of whale fall fossils of them. Our fossil record of whale fall communities really shows up when the big filter feeding whales come around. So 30 million years old is our oldest whale fall fossil, and that's when we start seeing the big filter feeding whales, the truly massive whales, blue whales, humpbacks, et cetera, and where their diversity really spikes.

And so really from the Elegocene onward is when we see significant whale fall fossils and, you know, the whale fall community really picked up. And while most whale fall fossils are because there's something growing on it, like a clam shoved up in the skull or something, we do get some that have mineralization on the skeleton that's evidence of those bacterial communities growing on the bones. There are pirateized skeletons that show evidence of possible microbial activity on the surface of the bone.

Evidence of stage three. Exactly. And so that's one thing we can also track is, you know, with the few fossils we have, how have the communities shifted or changed over time and what do we patterns do we see there? The other thing that I believe is the case for falls over time is that the organisms that make up the fall communities have themselves shown up and disappeared over time. Yes, absolutely. I remember reading about an ichthyosaur fall several years ago where, if I remember right, this has been several years, but the paper made the point that they didn't see evidence of those bone worms, the ocidax worms, because they hadn't evolved yet.

Yes. Those bone specialist worms, part of food falls for a long time because they didn't exist and that's a more recent thing. Yeah. And on that note, at least on one of the fossils, a Miocene whale fossil from Japan, it was noted that of the things on the skeleton, none of them were the same as the whale fall community members we see today. That makes sense. They had similar roles and organisms, but none were the same.

So there is a turnover. There has been changes. We still don't fully understand it because, as mentioned earlier, each individual whale fall is unique effectively, even today. It's hard to tell if the Miocene versus today difference is because of where it was in the world or what kind of whale it was or whatever, or if it is a time difference. If that

is the 10 million years or whatever that makes the difference. We also find fossils of vent communities, but also the communities on the vents.

So hydrothermal vents are very interesting as far as it goes with deep sea habitats because they are one of the earliest evidence of a habitat on earth. Vent fossils have been dated back, the oldest one I was able to find, back to 4.2 billion years old in the Hadean and has potential evidence of life on it. So that's one of the earliest examples of life we have, which makes it one of the earliest examples of a habitat we have.

So vents are very different than the other deep sea fossils we have. Now these oldest ones from the Precambrian, we have others that are younger than that, a number of vent fossils, not back to that age, but there is one from Canada dated between 3.7, 4.2 billion years old with micrometer hematite tubes, filaments on it that are similar in structure to vent life communities today. So that's the one that is pointed at as potentially the example of the earliest vent life community going back that far.

Gotcha. Which is interesting because, you know, we've discussed before early life, we've got fossils from around three and a half billion, and then things like that, occasional, maybe structural or chemical signatures of life, which is interesting because hydrothermal vents often do come up in discussions of what early, early, early life on earth may have looked like. Absolutely. And we will discuss that in this section. Oh, good. So even if those filaments are not life on that, it does mean that vent habitats have been around for that long.

Right. The real estate was available. Exactly. So it is one of the earliest places life could have been living. There are younger, you know, 1.4 billion year old fossils practically yesterday with more solid evidence of microbial communities, you know, more, more definite. And basically most of the vent fossils, at least that I found listed, have some evidence of at least microbial communities on them. So it seems like once vent communities were around, they've been pretty common on vents since then.

We have fossils ranging from the Precambrian, the Paleozoic, Mesozoic, and Cenozoic. So like all the way through, they're still not ridiculously common. We're not just finding truckloads of them usually, but they are notable. The fossils seem to indicate that by the Early Silurian or Ordovician, we see decent animal vent life community. So not just microbial, but actual animals growing on or swimming around it. There is some potential evidence for a sponge or maybe tube worm from the Cambrian, or I think a sponge and a tube worm fossils from the Cambrian, but those seem like they might be maybe not that.

Okay. So that would be about as early as you could get sponges and tube worms and vents. Exactly. But by the Ordovician, it seems like the animal vent-like community was there and was diverse. Cool. That's a fun thought because with the food fall communities, you couldn't get those before the Cambrian because there weren't big animals in existence. Exactly. You couldn't have a whale fall or even a fish fall, but vents have been around. It sounds like since about as early as the earth goes, so microbes could have been living on vents for billions and billions of years, even before animals ever showed up.

Which makes sense because the vents have, like a vent forming has nothing to do with life. Right. You can get a quote fossilized vent that isn't a fossil. It's just the shape. It's a geologic structure. Exactly. It's like a stalactite. Yes. It's kind of geologically giving us evidence of the shape of the sea floor, irrespective of whatever life was doing. It makes

perfect sense that that community has been ripe and ready for as long as life's been around.

Right. Now, the origin of vent bacteria in communities is extremely heavily debated. We don't know. There is some evidence that the bacteria on the vents, the sulfophilic bacteria didn't derive from photosynthetic, but evolved that way on its own. That you had life and it evolved, some of it evolved like sulfur, and then others evolved to like photosynthesizing. That life didn't like colonize the surface by being good at eating sunlight. And then some of them sprinkled down to the vents that very ancestral microbes gave rise to some descendants that were good at sulfur venti communities and some that went on to photosynthesizing.

Yes. There's also a lot of research on the relationship between food falls and vent communities as well as seep communities, cold seep communities, because a lot of the organisms that we see on a whale fall are similar to those we find on a vent and on cold seeps. When looking at just the general species of modern ones, there are seven groups shared between wood falls and whale falls, 11 between whale falls and hydrothermal vents, and up to 20 between whale falls and cold seeps.

So a lot of overlap between those communities. And of the bacteria that we find, there's been up to 190 different types identified on a single whale carcass, and about 20% of them are known from hydrothermal vents. Oh, interesting. So a lot of overlap between these communities, which has led to the hypothesis that whale falls act as the go-between. Yeah, that you're hopping around these points in the sea floor where there's enough nutrition to live. What they called evolutionary stepping stones to be able to bounce from one vent, because vents do come and go.

If that fissure in the earth closes up, your vent dies, but then that you can hop to the nearest whale fall, and then to the next whale fall, and then to the next vent, or the whale to wood to vent, or whale to seep, that you can bounce between these sulfophilic habitats. And if you're wondering, when we say they hop, obviously a clam is not crawling across the floor. No. But like you mentioned before, a lot of ocean animals disperse larvae.

Yes, exactly. Pump it into the water. They just go everywhere, and if one of them happens to find a whale fall, it can then sprout up there. It's kind of like plants send their seeds in all directions, and they can travel through the air, or on birds, or across ocean currents. And if some seeds wind up on an island, those plants can eventually colonize that island. This is like seeding the ocean and colonizing islands that come and go perpetually.

Yes. And there is conflicting evidence and support for this. Some molecular evidence seems to support that, yes indeed, these communities are connected. You know, genetic evidence. Whilst others, there's one study looking at RNA of mytilids, which is a little mollusk found on both whale falls and wood falls, as well as vent and seep communities. The RNA research linked that whale and wood falls were much more similar than they were to seep communities and vent communities. So it didn't seem to be as strong a connection, potentially.

There is also at least one whale fall fossil from the early Cenozoic, around the Oligocene, that seemed to lack the sulfide communities, that didn't have those sulfophilic bacteria or organisms that were growing on the bones. They were much more similar to today's woodfalls in their numbers and their makeup, which may have

meant that the sulfophilic stage, that final third stage of the whale fall, didn't come around until the early Miocene, like 11 million years ago, when we got big enough whales with enough oil and fat content to support them.

Oh, interesting. So that early whale falls may not have even been able to serve as stepping stones. For certain organisms that rely on those particular nutrient sources. Those sulfophilic groups, because those are the ones that we're really looking at. Sulfophilic ones are the ones that are found on all of them, basically. Right. Yeah, we understand how sharks get place to place. Yeah, that's easy. But yeah, the microbial communities are the things that are really digging in.

So there is some genetic and some fossil evidence that says, well, maybe today it could work that way, but it doesn't seem like that's the way they've been working. And there may not be that much community crossover as we think. So some evidence seems to support that there's a connection between whale falls and wood falls and vents and seeps, that those communities are kind of shared or traveling among all these different food sources. And some evidence that says maybe not as much.

There may be something else going on. Yeah. So lots of research into that. That is a still hotly being looked into aspect of these communities. Now the final thing I wanted to mention when it comes to deep sea ancient vent communities, and you touched on this when you mentioned the fact that they could very well be one of the earliest sites life occupied. There are a number of people hypothesizing that they were the first site occupied by life.

Yeah, we did talk about this a bit in episode 100 when we talked about the origins of life. Yes. So go check that out for a more in-depth discussion on the topic. But yeah, vents, deep sea hydrothermal vents have been pointed at as potentially the location, the type of habitat where life originated. We often hear the phrase, life came from the sea. And generally speaking, what people are referring to, we know that life existed in the ocean before it went anywhere else.

This is true for basically all major forms of life, bacteria and plants and animals, all of them have their ancestry from the sea. But that doesn't necessarily mean the depths, shallow oceans, light areas, places where you can photosynthesize all that are the sort of life rich sections of the ocean that we're often referring to. These hypotheses suggest that no, in fact, if you go back far enough, it might actually have originated way down where there wasn't sunlight to support it.

If we were magically able to just go and identify and find the first cell that ever was, that it would be on the side of a hydrothermal vent. Right. And now you mentioned those other habitats. There are plenty of hypotheses pointing to lakes and shores and shallow water for life to start. So it's not like this is the only one, but there are some potentially convincing lines of evidence pointing to vents to at least be a very ideal place.

Right. A candidate. Yeah. We don't have, oh yeah, this is the first cell. Definitely. This fossil, they've got the serial number is zero, zero, zero, zero, zero, zero one. Earth specimen number one. Yeah. We don't have definite. But there is some evidence that really seems like it could have been. One of these is just the fact that the microbes we find on hydrothermal vents today share a lot of features of what we expect very early life microbes to have been like, like just feature wise.

Right. The microbes that live in extreme environments today may have been quite well suited for the very extreme environment that was the early earth. Precisely. So it does

seem like there's probably, if not a close relation, a very close association between vent life community that we can observe and what earliest life was like. There's also the fact that, and we go into more detail on this in our origins of life episode, but one of the main stepping stones we have to figure out for life to form is for organic molecules to form.

We need to get organic structures that we can then make organisms out of. And typically those structures are made by organisms. That's where they come from today. But they had to come from somewhere else before they were organisms. So we look at, okay, what features, what functions, what processes on earth can form these things without life's help? And we found a number of them. Lightning strikes have been talked about, things like that. But hydrothermal vents are another place in these extreme heated environments that we have seen the potential for organic molecules to be formed spontaneously.

The earth can create the building blocks of life in places like that. Now I don't know that we've ever directly witnessed a hydrothermal vent do it for sure, but synthesized experiments recreating the situation, the attributes of a hydrothermal vent have successfully created organic molecules. They indicate that it's possible. Yes. I'm not even able to go so far to say that if vent communities are where life originated, it was probably more likely white smoker vent communities. Oh, instead of black smokers.

Yeah. That they have a more ideal set of minerals and chemicals going on to create the right molecules. So it might've been there. The logic at least lays out very well. We don't have the evidence to confirm it, but all of the pieces are there. And the way we see them today sure does seem to look like what we think it might have been like in those earliest days. So as we sort of over time have narrowed down our leading hypotheses for how life could have originated, hydrothermal vent communities of first life has remained one of the big topics that we keep talking about.

It also, and this is a different discussion, this is episode 26, but has implications for if we ever find hydrothermal vents active on other celestial bodies. Because we have found evidence of either vents or similar geological processes on other planets and moons in our solar system. Right. Because they can also be created by other means. There have been impact craters with evidence of similar vent like structures. Well, like you said, they tend to form in our oceans at spreading centers where the crust is nice and thin.

Well, another way to thin your crust is to throw a giant rock at it to put a dent in it. Yep. But, you know, we look at other planets or moons and we go, well, maybe life couldn't form there because it's so cold or because there's no atmosphere or whatever, whatever. But in places like, for example, Europa, it's not uncommon for people to go, well, at the surface. Yeah, absolutely. There's oceans there. Because a vent community has basically zero connection with the fact that there's a sun.

Right. And the thing that is perhaps above everything else so fascinating about the suggestion that life could have originated on Earth at deep sea habitats, completely cut off from surface stuff is that life today is 99.9% inseparable, tethered from the power of the sun. Life, as we know it, needs the sun. Yes. So the idea that that's an emergent quality of life, that it didn't actually start that way is a very, it's a very unintuitive idea.

And it also makes me think of when we talk about those groups of life where like, yeah, modern sloths are not a good representative of sloths through time because ancient slots were much different. So that's not a good example today of what they were like.

That all of life could potentially fit that category where we go, yeah, all life needs light. Well, today. Yeah. Yeah. You have a skewed view of it. Oh yeah, sure. You and your Phanerozoic.

Yeah. It's a pretty crazy concept. It almost adds to the alien aspect of how different the deep sea is to us and the fact that it could be the origin. So yeah, that's our discussion on the deep sea. There's a ton of things we didn't discuss. This is already going to very likely be a long episode after I finish up editing it. Deep sea is a big place. There's a lot of it, period, in all things that you could consider it with.

But if there is something we didn't discuss, feel free to let us know. If there's a favorite deep sea organism that didn't get mentioned, which is a bunch of them, let us know. I'd love to hear what people's favorite deep sea critters are. But before we wrap up, we have one last section, which is our patron question. At the end of the episodes, we like to answer a question asked by one of our patrons because if you are a patron with us on the Patreon at a certain level and up, you can submit questions that we will answer as part of the episodes.

What's our question for today? Today's question is not about the deep ocean, but it is ocean adjacent, literally. Okay. This question's from Sam. Sam asks, can you describe how a Mesozoic tide pool might look? Cool. Yeah. Mesozoic, of course, the Mesozoic era, the age of reptiles, 250 to 66 million years ago. Will, you're the ocean guy. Yes. So tide pools, tide pools are very simply areas along the shore at the beach where the tides come in and out when the water rise and lowers.

And when the tide goes out, when low tide hits, there are areas very often just because of rocks or the way the shore is shaped that pool water. Water gets trapped in these little pockets and these are called tide pools. And they're unique because they are ephemeral. They are only there during low tide. And then at high tide, they are just underwater, they're just ocean. They're just the beach now again, but the water part. But there are organisms that are actually kind of specialized to these tide pools that have adapted to be able to hide out when the water pulls away and survive further up shore away from the deeper water, or to be able to kind of thrive while they're isolated from everyone else.

So a lot of animals that will come out in these tide pools when the tide's away and all the bigger predators are gone and they can hang out in their little neighborhood. You see lots of organisms in there that if you've ever been to an aquarium with like touch areas, sea anemones, sea urchins, sea stars, sea cucumbers, crabs, other kelps and plant things like that, octopus have been known from tide pools. There are also sharks that specialize in tide pools, little tiny cat sharks and dog sharks.

There's that one that I think was in Blue Planet that can walk, can move from tide pool to tide pool by moving its pectoral fins. There's the hunting octopus that will run out of tide pools to grab crabs and stuff. As far as Mesozoic tide pools, a lot of the Mesozoic would probably look pretty similar to today. The Jurassic and Cretaceous, a lot of those groups were already pretty well established. We had familiar crustaceans and familiar echinoderms and things like that.

Most of the groups that you'd expect from a tide pool are much older than the Mesozoic, so they've probably been in tide pools for a long time. We probably would see a shift in the mid-ish Mesozoic because that's when the Mesozoic Marine Revolution happens, which we've mentioned a few times before. This is a time when we

see animals adapted to eating tough stuff increasing and armor and defenses increasing in ocean life as the dynamic between predator and prey shifted.

We see a lot more shelled organisms. We see increased shelliness on already shelled organisms. So that would probably show up over the course of the Mesozoic. Exactly. Depending on when it happened, from what I could find, there's debate as to whether it started around the late Triassic or late Cretaceous was when it originally was dated. So somewhere in there, you would see a shift. I think it's safe to say, for example, that an early Triassic tide pool would be much more boring on the opposite end of the worst mass extinction of the Phanerozoic.

Probably not a lot of things thriving in the earliest days of the Triassic. And even throughout the Triassic, there are a number of groups that don't show up until after the Triassic. We don't get true crabs until after the Triassic, so you could have crustaceans, or crab-like things, sea crabs, pseudo crabs, imitation crabs, but not true crabs. You also would have likely had more crinoids. Oh, sure. Yeah, yeah. Sea lilies. Yeah. So crinoids, we see their numbers dip after the Triassic, especially for those close to shore varieties, but they were more common.

So I would be surprised if there weren't tide pool sea lilies. Cool. Octopuses also don't show up until the mid-Jurassic. So you wouldn't have -- you might have -- I would not be surprised if there were shallow-adapted cephalopods with shells. But I don't know -- Yeah, like tide pool ammonites. Yeah, I don't know if there would be tide pool -- I'd be surprised, once again, if there weren't ammonites specialized for shallow water that did their spiral sideways.

But octopuses wouldn't have shown up until mid-Mesozoic. The other interesting thing to think of with a Mesozoic tide pool is that you do get outside visitors. Right. Birds and, like, shoreline animal -- like bears that visit the beach -- have been known to go and eat the mussels and animals in tide pools. So I can definitely picture those being replaced by dinosaurs and pterosaurs through the Mesozoic. Especially little dinosaurs that could be picking on the stuff in there, or pterosaurs, absolutely.

Especially since we so often reconstruct them as feeding near the water. Yes. Because we find a lot of them near the water. But yeah, pterosaurs acting like gulls, pecking around in tide pools. So I'd say probably they'd be pretty -- like, you might be able to hold up a Mesozoic tide pool and a today tide pool and not be able to immediately recognize which was which. But yeah, there would be some significant differences, and we would see a shift through the Mesozoic to a more recognizable set of tide pools.

I wonder if -- and I can't think of an animal today that would -- like, do baby fish hang out in tide pools? I do know that there are some that'll do stuff like that, but I don't know if it's as regular as, like, ones that use the mouths of rivers and stuff. Because tide pools are -- at least to my understanding -- can be very inconsistent. Like if you were created -- if a tide pool was created because of the way the rocks were there, if those rocks shift, that tide pool might drain the next time.

But maybe every now and then I'd imagine a bigger fish might get caught in a tide pool, so you might find something -- I'm trying to come up with a way that you could get a baby mosasaur in a tide pool that just every now -- like, a particularly large tide pool and just one newborn mosasaur that just happened to be swimming around when the tide went out and is now -- for several hours -- the worst nightmare of every other thing in that tide pool.

The community would talk about it generations later, "Oh, yes, grandpa, my shrimp grandfather tells me of the time when the beast..." There were small mosasaurs, right? Like -- I mean, there were mosasaurs that were, you know, a few feet long. Yeah. So, I mean, like, those would have pretty tiny babies. Yeah, absolutely. Like, those could fit in there. There had to have been -- like, there are sharks. Small ichthyosaurs and plesiosaurs. There are sharks today that are specialized for tide pools.

There had to have been other marine predators that were specialized for tide pools that were itty-bitty. So, yeah. Yeah. Good question. Very cool question. And with that, we can wrap up this episode. We can surface. We can surface. Slowly. Slowly. Yeah, right, right, right. There was a lot to discuss in the deep sea topic, so, as usual, if there is something we didn't discuss or a favorite thing that you wished we had said that we didn't, let us know.

You can request episode topics like the people who did this one, like all of our episodes -- social media, email, blog, however you want to reach out to us. You can mail it to us if you want now. Yeah, you can put it in a letter. There will be a blog post about this topic, as usual, with links and pictures, too, if you want to dive even deeper into the subject. Go check the Jurassic World Dominion promo silver screen science thing we did.

Absolutely. And if you're listening to this in the first couple days of this episode's release, you might still be able to submit a question for our end-of-the-year Q&A. And then be able to keep an ear out for that end-of-the-year Q&A at the end of the year. At the end of the year, right where it goes. And as usual, we release episodes every fortnight. So stay tuned. There's one more. There's one more. One more for 2021 before we move on into a whole new year.

Yeah. So we'll see you then. A ta-ta for now. Thanks again for listening. We hope you'll join us next time.